

Construct a modular Embedded C program using multiple files for a multi-sensor system and analyze scalability.

PROGRAM:

```
#include <stdio.h>

/===== STRUCTURE FOR MULTI-
SENSOR DATA =====/

typedef struct { float temperature; int heartRate; int spo2; } SensorData;

/===== TEMPERATURE SENSOR
MODULE =====/

void Temperature_Init(void) { printf("Temperature Sensor Initialized\n"); }

float Temperature_Read(void) { return 36.8f; // Simulated temperature }

/===== HEART RATE SENSOR
MODULE =====/

void HeartRate_Init(void) { printf("Heart Rate Sensor Initialized\n"); }

int HeartRate_Read(void) { return 75; // Simulated heart rate }

/===== SPO2 SENSOR MODULE
=====/

void SpO2_Init(void) { printf("SpO2 Sensor Initialized\n"); }

int SpO2_Read(void) { return 97; // Simulated SpO2 value }

/===== SENSOR MANAGER
MODULE =====/

void Sensors_Init(void) { Temperature_Init(); HeartRate_Init(); SpO2_Init(); }

void Sensors_ReadAll(SensorData *data) { if (data == NULL) return;

data->temperature = Temperature_Read();
data->heartRate = HeartRate_Read();
```

```

data->spo2 = SpO2_Read();

}

/===== DISPLAY MODULE
=====

void Display_Init(void) { printf("Display Initialized\n"); }

void Display_Data(SensorData *data) { printf("\n-----\n"); printf("
SENSOR DATA\n"); printf("-----\n");

printf("Temperature : %.2f C\n",
        data->temperature);

printf("Heart Rate   : %d bpm\n",
        data->heartRate);

printf("SpO2          : %d %%\n",
        data->spo2);

printf("-----\n");

}

/===== PATIENT STATUS MODULE
=====

void CheckPatientStatus(SensorData *data) { if (data->heartRate < 40 || data->heartRate >
120) { printf("WARNING: Abnormal Heart Rate!\n"); }

if (data->temperature < 35.0 ||
    data->temperature > 38.5)
{
    printf("WARNING: Abnormal Temperature!\n");
}

if (data->spo2 < 90)
{

```

```

        printf("WARNING: Low SpO2!\n");
    }

    if (data->heartRate >= 40 &&
        data->heartRate <= 120 &&
        data->temperature >= 35.0 &&
        data->temperature <= 38.5 &&
        data->spo2 >= 90)
    {
        printf("Patient Status: NORMAL\n");
    }

}

/===== MAIN PROGRAM
=====/

int main(void) { SensorData data;

    printf("=====\n");
    printf(" MULTI-SENSOR HEALTHCARE SYSTEM\n");
    printf("=====\n");

    /* Initialize all sensors */
    Sensors_Init();

    /* Initialize display */
    Display_Init();

    /* Read sensor values */
    Sensors_ReadAll(&data);

    /* Display sensor data */
    Display_Data(&data);

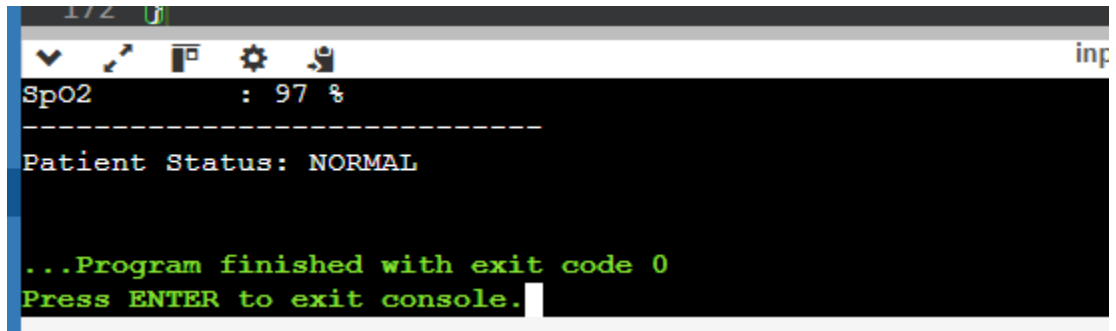
    /* Check patient condition */
    CheckPatientStatus(&data);

```

```
return 0;
```

```
}
```

OUTPUT:

A screenshot of a terminal window with a dark background. The window title bar at the top shows '1/2' and a green icon. Below the title bar is a toolbar with icons for a dropdown menu, a cursor, a window, a gear, and a document. The text 'inp' is visible on the right side of the toolbar. The main area of the terminal displays the following text in white: 'SpO2 : 97 %', followed by a dashed line '-----', and then 'Patient Status: NORMAL'. At the bottom, green text reads '...Program finished with exit code 0' and 'Press ENTER to exit console.' with a white cursor at the end of the second line.

```
1/2 [icon]  
inp  
SpO2 : 97 %  
-----  
Patient Status: NORMAL  
...Program finished with exit code 0  
Press ENTER to exit console.
```